Derivation of the Wasserstein Gradient Flow Equation

In Euclidean space, the gradient flow of a function f is

$$\dot{x}_t = -\nabla f(x_t).$$

This says that the point moves in the direction of steepest decrease, measured using the Euclidean inner product.

In Wasserstein space, points are probability densities ρ_t. A curve of densities moves through a velocity field v_t satisfying the continuity equation:

$$\partial_t \rho_t + \operatorname{div}(\rho_t v_t) = 0.$$

The Wasserstein metric measures the size of the velocity by kinetic energy:

$$\|v_t\|_{\rho_t}^2 = \int_{\mathbb{R}^d} |v_t(x)|^2 \rho_t(x)\, dx.$$

Derivative of the Functional

Let $\mathcal{F}$ be a functional on densities. Along a curve ρ_t,

$$\frac{d}{dt}\mathcal{F}(\rho_t)$$

means the rate at which the number $\mathcal{F}(\rho_t)$ changes as the density ρ_t moves.

The first variation plays the role of a gradient. In finite dimensions, if $f : \mathbb{R}^n \to \mathbb{R}$ and x_t is a curve, then the chain rule gives

$$\frac{d}{dt} f(x_t) = \nabla f(x_t) \cdot \dot{x}_t = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(x_t)\dot{x}_i(t).$$

For a density ρ_t, the "coordinates" are the values $\rho_t(x)$ indexed by the continuous variable $x \in \mathbb{R}^d$. Thus the finite-dimensional sum is replaced by an integral:

$$\sum_i \frac{\partial f}{\partial x_i}\dot{x}_i \quad \rightsquigarrow \quad \int_{\mathbb{R}^d} \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t)(x)\partial_t \rho_t(x)\, dx.$$

Here:

- $\partial_t \rho_t(x)$ is the infinitesimal change of the density at x,
- $\frac{\delta \mathcal{F}}{\delta \rho}(\rho_t)(x)$ is the sensitivity of $\mathcal{F}$ to changing the density at x.

Therefore the functional chain rule is

$$\frac{d}{dt}\mathcal{F}(\rho_t) = \int_{\mathbb{R}^d} \frac{\delta \mathcal{F}}{\delta \rho}(\rho_t)(x)\partial_t \rho_t(x)\, dx.$$

For example, if

$$\mathcal{F}(\rho) = \int_{\mathbb{R}^d} V(x)\rho(x)\,dx,$$

then

$$\mathcal{F}(\rho_t) = \int_{\mathbb{R}^d} V(x)\rho_t(x)\,dx,$$

so direct differentiation gives

$$\frac{d}{dt}\mathcal{F}(\rho_t) = \int_{\mathbb{R}^d} V(x)\partial_t\rho_t(x)\,dx.$$

This matches the general formula because

$$\frac{\delta\mathcal{F}}{\delta\rho} = V.$$

Using the continuity equation,

$$\frac{d}{dt}\mathcal{F}(\rho_t) = -\int_{\mathbb{R}^d} \frac{\delta\mathcal{F}}{\delta\rho}(\rho_t)\,\mathrm{div}(\rho_t v_t)\,dx.$$

Integrating by parts gives

$$\frac{d}{dt}\mathcal{F}(\rho_t) = \int_{\mathbb{R}^d} \nabla\left[\frac{\delta\mathcal{F}}{\delta\rho}(\rho_t)\right]\cdot v_t\,\rho_t\,dx.$$

Identifying the Wasserstein Gradient

The Wasserstein inner product of two velocity fields is

$$\langle u, v\rangle_{\rho_t} = \int_{\mathbb{R}^d} u\cdot v\,\rho_t\,dx.$$

The identity obtained above was

$$\frac{d}{dt}\mathcal{F}(\rho_t) = \int_{\mathbb{R}^d} \nabla\left[\frac{\delta\mathcal{F}}{\delta\rho}(\rho_t)\right]\cdot v_t\,\rho_t\,dx.$$

Using the Wasserstein inner product, this can be rewritten as

$$\frac{d}{dt}\mathcal{F}(\rho_t) = \left\langle \nabla\left[\frac{\delta\mathcal{F}}{\delta\rho}(\rho_t)\right], v_t\right\rangle_{\rho_t}.$$

This is the direct analogue of the Euclidean identity

$$\frac{d}{dt}f(x_t) = \nabla f(x_t)\cdot\dot{x}_t.$$

In Riemannian geometry, the gradient is the vector that represents the directional derivative through the metric:

$$D\mathcal{F}(\rho_t)[v_t] = \langle \mathrm{grad}_{W_2}\mathcal{F}(\rho_t), v_t\rangle_{\rho_t}.$$

Comparing this definition with the previous formula, the velocity-field representative of the Wasserstein gradient must be

$$\nabla\left[\frac{\delta\mathcal{F}}{\delta\rho}(\rho_t)\right].$$

Therefore the negative-gradient velocity is

$$v_t = -\nabla\left[\frac{\delta\mathcal{F}}{\delta\rho}(\rho_t)\right].$$

Plugging Into the Continuity Equation

Substituting this velocity into

$$\partial_t \rho_t + \mathrm{div}(\rho_t v_t) = 0$$

gives

$$\partial_t \rho_t = \mathrm{div}\left(\rho_t \nabla \left[\frac{\delta \mathcal{F}}{\delta \rho}(\rho_t)\right]\right).$$

This is the Wasserstein gradient flow equation.